

ForecastIQ

Probabilistic revenue planning and risk-aware budget guidance — 30, 60, and 90-day outlook

AUDITDATE	DATAQUALITY	RISKCLASSIFICATION	PLATFORMMODE
2026-07-17	100.0/100	High Risk	Offline Utility

1 Executive Summary

KEY TAKEAWAY

Over the full historical timeline, ForecastIQ has audited \$2.18M in digital marketing spend yielding \$11.10M in revenue (Portfolio ROAS: 5.09x). Our 90-day probabilistic ensemble projects sustained top-line stability with a P50 expected capture of \$0.43M. Meta Ads continues to function as your high-efficiency anchor at 8.44x ROAS, while Google Ads shows the strongest recent spend momentum for scalable incremental demand.

2 Multi-Horizon Revenue & ROAS Projections

The following forecasts are derived from a weighted ensemble (Prophet, XGBoost, LightGBM, CatBoost). P10 is a conservative floor, P50 the expected baseline, and P90 an upside scenario — not point estimates, but a probability range meant to be planned against.

Planning Window	Metric	P10 (Conservative)	P50 (Expected)	P90 (Upside)
30 Days	Revenue	\$67,764	\$141,880	\$215,996
30 Days	ROAS	1.16x	2.42x	3.68x
60 Days	Revenue	\$159,980	\$284,628	\$409,276
60 Days	ROAS	1.35x	2.40x	3.45x
90 Days	Revenue	\$258,910	\$427,858	\$596,806
90 Days	ROAS	1.45x	2.40x	3.35x

3 Forecast Accuracy: Offline Backtest vs. Naive Baseline

3-fold rolling-origin

To check whether the uncertainty bands above are meaningful rather than arbitrary, the ensemble was evaluated with a rolling-origin backtest: retrained on progressively larger historical windows and scored against forecast periods it had not seen, then compared to a naive baseline (a flat projection of the trailing 30-day daily average). Figures below are computed at the Overall (portfolio) level, where percentage-based accuracy is stable; Campaign-level rows are noisier and are addressed separately below.

Window	Metric	Model MAPE	Naive Baseline MAPE	P10–P90 Coverage	Result
30 Days	ROAS	20.0%	23.8%	67%	Beats baseline (+16.1%)
30 Days	Revenue	18.5%	17.6%	33%	Trails baseline (-5.1%)
60 Days	ROAS	9.4%	13.3%	67%	Beats baseline (+29.4%)
60 Days	Revenue	17.4%	18.4%	33%	Beats baseline (+5.5%)
90 Days	ROAS	16.8%	18.0%	33%	Beats baseline (+6.7%)
90 Days	Revenue	6.1%	6.6%	67%	Beats baseline (+6.9%)

ROAS forecasts beat the naive baseline at all three horizons (30-day, 60-day, 90-day). Revenue forecasts beat the naive baseline at 60-day (+5.5%), 90-day (+6.9%) and trail it at 30-day (-5.1%). We're disclosing this rather than omitting it. Tracing the gap by decomposing the forecast into its components found that the model's own recency-anchoring step — which blends the raw ensemble output with a trailing revenue baseline to stabilize it — was weighting a stale 90-day average too heavily relative to the more recent 30-day level the naive baseline itself uses, overshooting by 18–23% APE on its own before the underlying model even contributed. Re-weighting that anchor toward the same recency the naive baseline uses cut the Revenue MAPE gap by roughly 60–75% across all three horizons. Two further factors widen the remaining gap: (1) the earliest backtest origin trains on only ≈ 4 months of history — well short of what Prophet's seasonal component needs to calibrate reliably, and (2) the dataset contains a genuine 5–10x November–December demand spike that appears in only two historical instances across the ≈ 2.5 -year dataset, so any forecast horizon crossing that window is working with minimal precedent — a constraint that affects any forecasting method, not just this one.

KNOWN LIMITATIONS

Forecast accuracy improves with training history and is weakest for the earliest, data-sparse backtest origin. November–December forecasts carry wider uncertainty given only two observed instances of the holiday demand spike in the historical record. Campaign-level forecasts (excluded from the table above) inherit materially higher variance than portfolio-level ones, since individual campaign lifecycles and end-dates aren't modeled as an explicit signal.

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Channel Performance & Contribution Breakdown

Treating existing channel attribution as the source of truth, here is the forward 60-day expected contribution and efficiency analysis across your primary paid acquisition channels.

Acquisition Channel	Metric	P10	P50 (Expected)	P90
Bing Ads	Revenue	\$228	\$4,560	\$16,968
Bing Ads	ROAS	0.03x	0.67x	2.51x
Google Ads	Revenue	\$328,389	\$572,625	\$816,860
Google Ads	ROAS	3.25x	5.66x	8.07x
Meta Ads	Revenue	\$1,174	\$23,485	\$132,601
Meta Ads	ROAS	0.08x	1.56x	8.80x

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Budget Optimizer Recommendations

An Optuna-driven search allocated a monthly budget of **\$100,000** to maximize total revenue subject to a target ROAS floor of **4.5x**. The search assumes diminishing marginal returns per channel — pushing more spend into one channel yields progressively less revenue per dollar — so the optimizer balances the split across all channels rather than concentrating on whichever looks best in isolation.

Channel	Recommended Spend	Expected Revenue	Expected ROAS	Spend Share
Bing Ads	\$6,653	\$20,967	3.15x	6.7%
Google Ads	\$55,988	\$272,546	4.87x	56.0%
Meta Ads	\$37,346	\$245,554	6.58x	37.4%
OPTIMIZED TOTAL	\$99,987	\$539,067	5.39x	100.0%

READING THIS TABLE

Google Ads receives the largest share of this recommendation (56% of budget) at an expected 4.87x ROAS — as spend rises, revenue increases but marginal ROAS typically falls. The optimizer's job is to find the allocation that protects your ROAS floor while still increasing revenue — a business decision, not just a bigger number.

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Risk Intelligence & Growth Action Plan

The overall portfolio risk classification is **High Risk** (risk score: 70.3/100). Below are the specific operational risks identified and recommended mitigation actions.

Risk Factor	Score	Status	Mitigation
Revenue Volatility	100.0/100	Monitor	Utilize automated pacing tools to smooth out mid-week revenue depressions and maintain consistent top-of-funnel impression share.
Channel Dependency	58.5/100	Concentrated	Google Ads represents 83.5% of revenue capture. Begin scaling top-of-funnel prospecting on your other channels to establish a strong secondary growth pillar.
ROAS Instability	85.4/100	Volatile	Enforce rigorous Target CPA / Target ROAS bid floor rules across all Bing Ads generic and non-brand campaigns to prevent runaway spend.
Data Quality Audit	0.0/100	Excellent	Validation engine verified a 100.0/100 schema score. Ensure UTM tracking tags are consistently and correctly appended across all ingested channel campaigns.

ForecastIQ — built for the NetElixir AIgnition 3.0 Hackathon Challenge.

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